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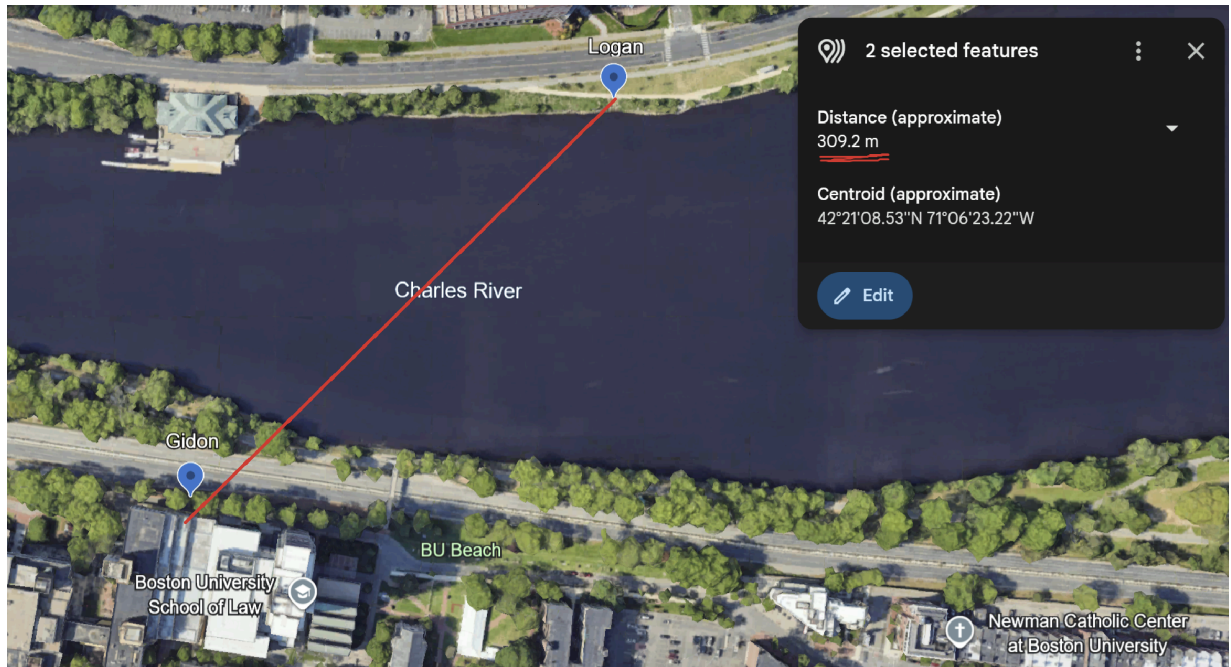
This document contains data and analysis from two range tests, along with a comparison between the results of the two tests. One test is of a much shorter distance (300m) and one is a much longer range (1600m) test. The data comes from tests run by electrical lead *Gidon Gautel* along with the communications Team: *Logan Fick*, *Jason Nguyen*, and *Stanley Lin*.

In these tests, the base station is the RocketM2 and the antenna is the Bullet Ip67, both from Ubiquiti. The frequency being used is 2.4ghz with automatic channel selection and swapping enabled. The information for these devices along with other network information and firmware documentation can be found in the [Communications Team Drive](#).

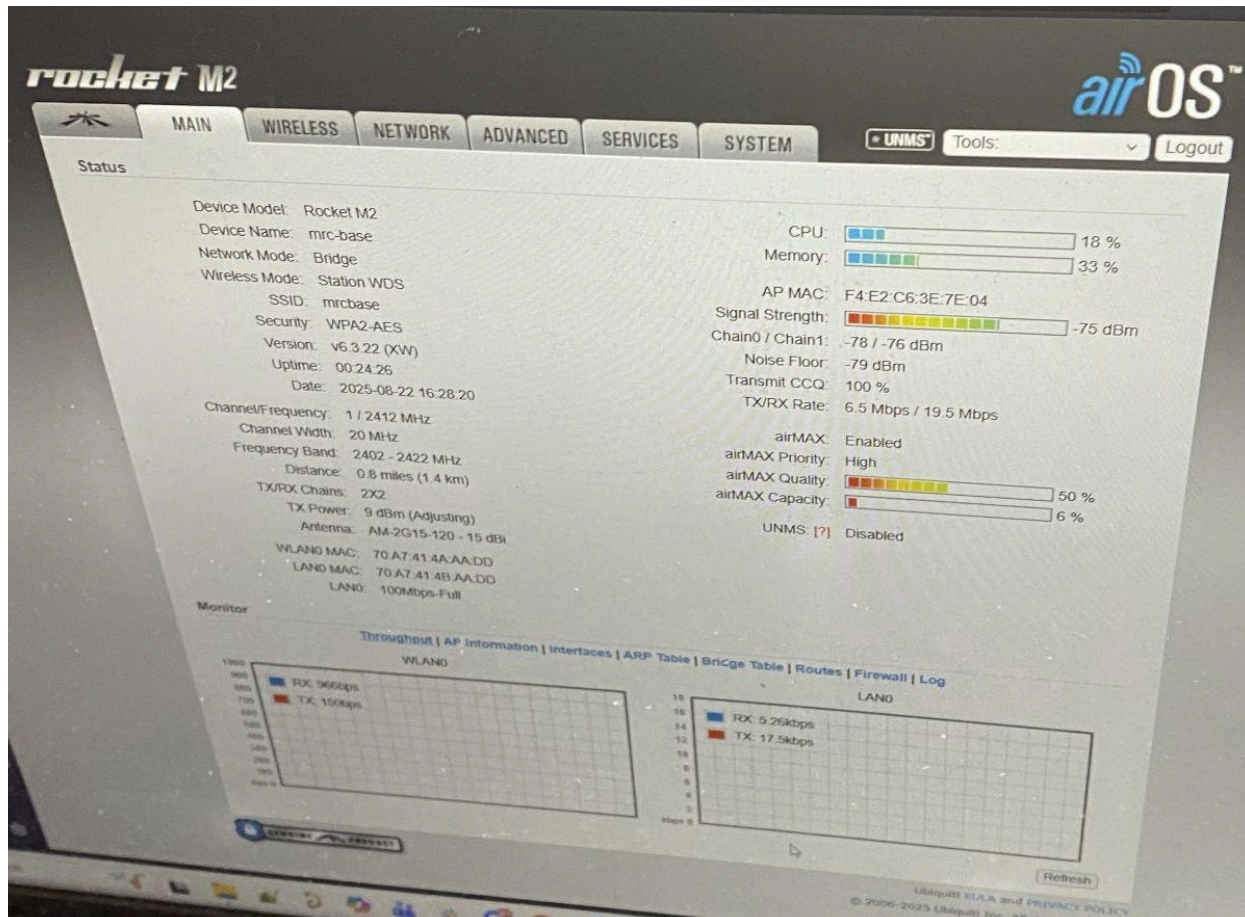
RANGE TEST ONE (11/6/2025)

APPROXIMATE DISTANCE 0.31KM (~0.19 MILES)

WITH LINE OF SIGHT THROUGH A GLASS WINDOW



RESULTS FOR RANGE TEST ONE



DISCUSSION OF RESULTS FOR RANGE TEST ONE

Above is a photo taken from Gidon's laptop while he was connected to the RocketM2 Web Panel and I (logan) was 309m (0.19mi) away across the Charles River. The connection is somewhat strong with the signal strength being -75dBm and the noise floor being -79dBm.

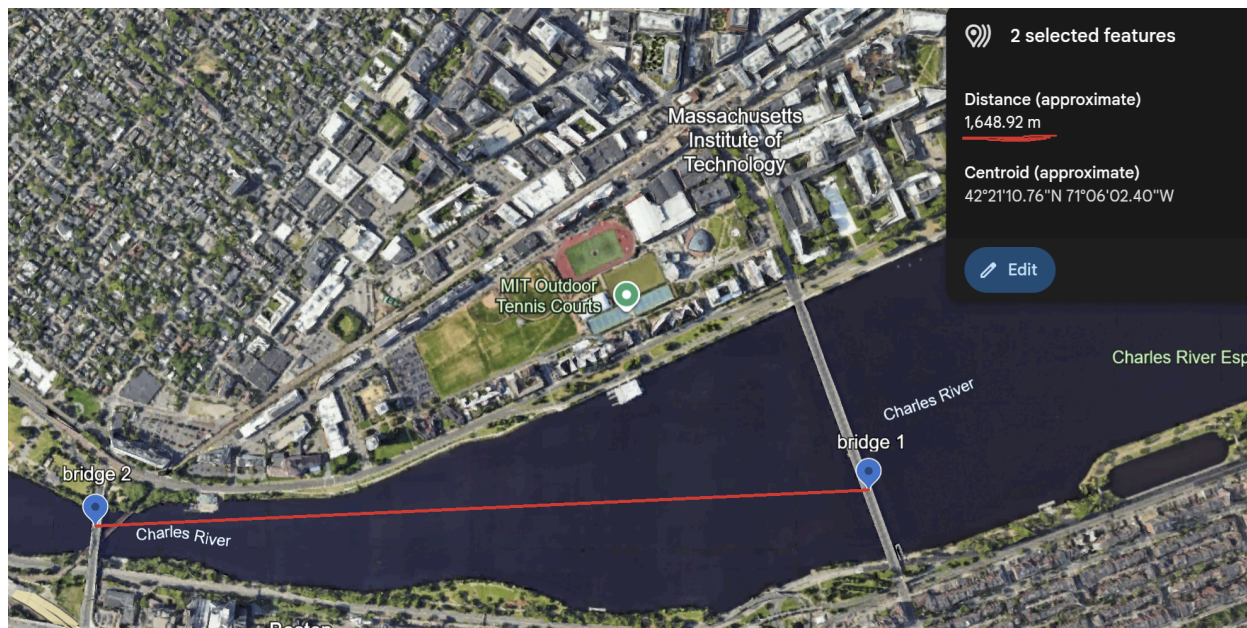
The airMAX quality is 50% indicating a usable connection and the RX rate is 19.5 Mbps with a TX rate of 6.5 Mbps.

It seems that this connection is quite usable for sending and receiving small to medium amounts of data with the rover. Without details of the video stream it is hard to tell the possible framerate or video quality between the camera and the base station.

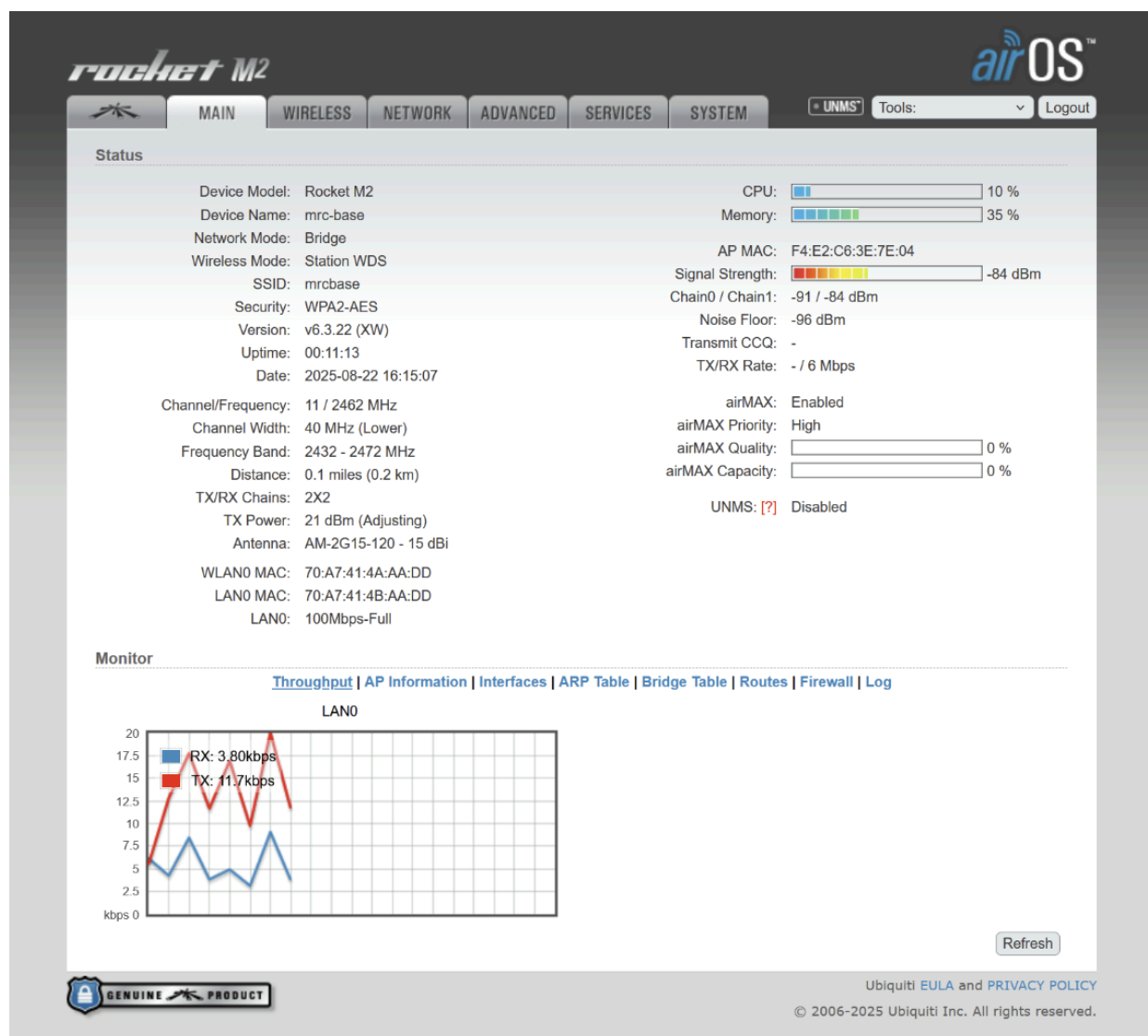
RANGE TEST TWO (2/5/2026)

APPROXIMATE DISTANCE 1.65KM (~0.99 MILES)

WITH LINE OF SIGHT



RESULTS FOR RANGE TEST TWO



DISCUSSION OF RESULTS FOR RANGE TEST TWO

Above is a screenshot from the RocketM2 web panel displaying a connection between the two points on the map being made. It appears that the Rocket was able to transmit between the Bullet and with an approximate RX rate of 6Mbps and an undefined TX rate.

The signal strength was -84dBm with the noise floor being -96dBm, meaning the signal was just barely above the background noise of the environment. However, the rover will not need to significantly go further than this distance I believe.

Possible reasons for this outcome is that the test was executed in a city with a lot of interference. The 2.4ghz channel is particularly crowded in an urban environment as well. So, it appears that with a connection made in such a crowded and noisy environment, the real communications environment will allow for a much stronger connection. A possible reason for the undefined TX rate could be an unstable signal that was unable to properly determine its TX rate. This could be because of the aforementioned reasons for a weaker signal.

COMPARING RESULTS BETWEEN THE TWO TESTS

The second test had a significantly weaker signal (-75dBm -> -84dBm) but that was as a result of increasing the range by a factor of over 5 times (300m -> 1600m). The noise floor actually fell in the second test (-79dBm -> -96dBm) and this could be a result of being further from buildings since the first test was conducted from inside mugar to the river and the second test was bridge to bridge.

The second test correctly increased the TX power from 9dBm to 21dBm as a result of the increased distance. The tests used different 2.4Ghz channels which indicates the automatic channel swapping and selection is functioning as expected.

Overall the distance estimate seems to be very incorrect.

The TX/RX speed was not very easy to discern for the second test but there was a reading of 6Mbps on the RX rate which is down from 19.5Mbps in the first test.